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Seeing the Invisible: Grad-CAM-Driven Detection of Sparse and Imperceptible Backdoor Attacks

***Abstract*—The reliability of deep neural networks is seriously threatened by sparse and invisible backdoor attacks. Sparse and invisible backdoor attacks are difficult to detect because their triggers cannot be seen by humans. In this paper, we propose an explainable backdoor detection method based on comparing model behavior using Grad-CAM. The method measures differences in attention patterns between a clean model and a suspicious model for clean and triggered inputs. Simple features are extracted from these differences and used to train different classifiers for detection. Experiments on CIFAR-10 with SIBA backdoor attacks and different classifiers show that the proposed method achieves 98.34% accuracy and a ROC-AUC of 99.32% for the spatial attention classifier. The results demonstrate that explanation-based model comparison is an effective and interpretable way to detect sparse and invisible backdoor attacks.**

***Index Terms*—Backdoor attacks, Sparse and invisible triggers, Model security, Explainable AI, Grad-CAM, Backdoor detection, Deep neural networks, Spatial attention.**

I. Introduction

Deep neural networks are frequently used in tasks including image classification, facial recognition, and autonomous systems. But because of backdoor attacks, their widespread use makes security a big worry. An attacker poisons a small part of the training data in these kinds of attacks. When there is a hidden trigger, the model works fine with clean inputs but gives wrong outputs. This kind of behavior is hard to spot and can be dangerous in applications where safety is very important.

Many backdoor attacks have been proposed, including visible patch-based triggers and invisible perturbations [1], [2] . To defend against these attacks, researchers have developed various detection and defense methods, such as model analysis, test-time detection, and training-data filtering. However, many existing defenses rely on assumptions such as trigger visibility, spatial localization, or strong activation patterns, which do not hold for more advanced attacks.

A recent and challenging example is the *Sparse and Invisible Backdoor Attack (SIBA)* [3]. SIBA modifies only a few pixels (50 pixels and about 5% sparsity) with very small perturbations and embeds the trigger within meaningful image regions. As a result, it achieves high attack success rates while bypassing many state-of-the-art defenses, including Neural Cleanse, STRIP, Anti-Backdoor Learning, fine-pruning, and explainability-based methods such as SentiNet [4]. Notably, Grad-CAM visualizations of clean and backdoored inputs appear almost identical, making visual inspection ineffective.

To understand how models make decisions, a lot of people use Grad-CAM and other explainability methods [5], [6]. However, most explainability-based defenses treat Grad-CAM mainly as a visual localization tool and assume that backdoor triggers produce clear and localized attention patterns. This assumption fails for sparse and invisible attacks like SIBA, highlighting the need for new detection approaches.

In this work, we revisit Grad-CAM from a new perspective. Instead of using it for direct trigger localization, we treat Grad-CAM as a behavioral signal and compare attention patterns between a clean reference model and a suspected backdoored model. We observe that for triggered inputs, SIBA causes clear differences in attention behavior across models, even though the saliency maps within a single model appear similar.

Based on this observation, we detect SIBA-style backdoors by extracting simple statistical and similarity-based features from Grad-CAM heatmap differences and applying supervised classification.

The main contributions of this paper are summarized as follows:

1) We propose a Grad-CAM–based model comparison method for detecting sparse and invisible backdoor attacks (SIBA).
2) The method finds backdoors by looking at the differences in attention between clean and suspected models.
3) We make a small, easy-to-understand set of features from the differences in Grad-CAM heatmaps.
4) We show that the proposed method works well by testing it on the CIFAR-10 dataset.

II. Literature Review

Backdoor attacks expose deep neural networks in danger by inserting malicious behavior into the training process without anyone knowing it. In these attacks, models act normally when they get clean inputs, but when a trigger is present, they give outputs that the attacker chooses. Initial research, including BadNets, showcased discernible patch-based backdoors [7]. Later studies introduced more covert triggers, including blended patterns [8], warping-based triggers [9], and imperceptible perturbations [10].

Most current attacks achieve either sparsity or invisibility, but not concurrently [11], [12], [12]–[19]. Gao et al. recently brought forward the Sparse and Invisible Backdoor Attack (SIBA), which uses bi-level optimization to make sure both properties hold true [3]. SIBA only changes a few pixels by a very small amount, and it often lines up the trigger with semantic regions, which makes it hard to find.

There are three types of backdoor defenses: input-level, training-level, and model-level. Input-level methods, like

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STRIP [20] and Scale-Up [21], find poisoned samples during testing, but they don't work on sparse or hidden triggers. Training-level defenses like ABL try to get rid of poisoned samples during training [22], but they don't work well against SIBA because they lose in the same way. Model-level defenses, such as Neural Cleanse [23] and Beatrix [24], are only effective against attacks that are easy to see or understand.

Recently, explainability techniques like Grad-CAM have been investigated for backdoor detection [5], [6], [25]. Grad-CAM highlights important image regions for model predictions, but it fails to detect SIBA because the triggers are extremely sparse and embedded within semantic regions. As a result, saliency maps of clean and poisoned inputs appear nearly identical, limiting the effectiveness of existing explainability-based defenses.

III. Methodology

A. Threat Model and Assumptions

This work considers a forensic backdoor detection setting, where a suspicious model is compared with a trusted clean model trained on the same task. We assume white-box access to the models to extract Grad-CAM explanations. Known triggers are used only for evaluation, and the detector itself does not rely on trigger knowledge. To avoid label mismatch, Grad-CAM heatmaps are computed using the same class index (ground-truth label) for both models. Although detection is performed per input, a model is flagged as backdoored if abnormal behavior is observed for a sufficient number of clean test inputs.

B. Overview of the Detection Pipeline

Figure 1 shows the whole SIBA backdoor detection pipeline. The pipeline shows how Grad-CAM heatmaps are generated from clean and backdoored models, how heatmap differences are computed, and how features are extracted and classified using different classifiers.

C. Data Collection

1) Dataset: All experiments are conducted on the CIFAR-10 dataset, which contains 60,000 RGB images of size 32×32 from 10 object classes. We use the standard training set of 50,000 images for model training and the official test set of 10,000 images for backdoor detection and evaluation.

2) Backdoored Model Generation: To obtain a compromised model, we generate a backdoored neural network following the Sparse and Invisible Backdoor Attack (SIBA) [3]. SIBA embeds a trigger that is both sparse and low in magnitude, making poisoned samples visually indistinguishable from clean images.

A small part of the training data is poisoned during training by adding the SIBA trigger and putting these samples in a target class that has already been set. The rest of the training samples are kept clean. This process makes a model that has a backdoor and works normally on clean inputs but misclassifies inputs that trigger it.

3) Model Training: Two models with the same architectures and training settings are trained:

1) a **clean (benign) model**, trained only on clean CIFAR-10 training data, and
2) a **backdoored model**, trained on a mixture of clean and SIBA-poisoned training samples.

These two models form the basis of our comparative detection framework.

4) Test-Time Input Preparation: For detection, we use only the CIFAR-10 test set. For each test image, we generate two versions:

- a clean test image, and
- a triggered test image created by applying the same SIBA trigger.

No test images are used during training.

5) Grad-CAM Heatmap Generation: The clean model and the backdoored model are both used on both the clean and triggered versions of each test image. The last convolutional layer of each model is used to make Grad-CAM heatmaps.

This results in four heatmaps per test image:

1) clean model on clean image,
2) clean model on triggered image,
3) backdoored model on clean image, and
4) backdoored model on triggered image.

These heatmaps let us directly compare the attention behavior of clean and backdoored models, which is the basis of our detection method.

D. Grad-CAM-Based Analysis

To get class-discriminative attention maps, Grad-CAM is used on the last convolutional layer of each model. Both models should focus on areas that are important to the object when the images are clean. On triggered images, the backdoored model shows strange changes in attention because of the backdoor that was added, while the clean model keeps stable attention patterns. Figure IV-C

E. Heatmap Difference Computation

For each input image, we compute the difference between the Grad-CAM heatmaps generated by the clean model and the suspected model. This difference highlights changes in both the location and strength of model attention.

$$D = H_s - H_c \tag{1}$$

In this case, H_s is the Grad-CAM heatmap from the model that is thought to be backdoored, H_c is the heatmap from the clean model, and D is the attention difference map. The values in D are usually small for clean images, but they are bigger and more structured for triggered images, which could mean that a backdoor was activated.

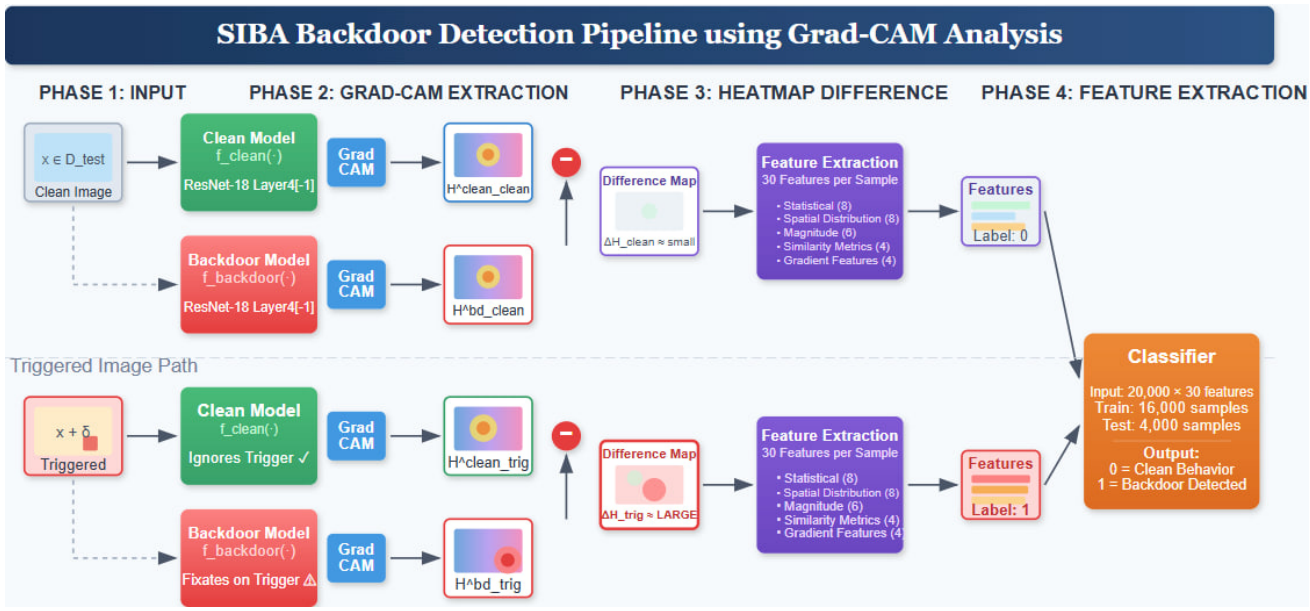

Fig. 1: Overview of the proposed SIBA backdoor detection pipeline. Clean and triggered images are processed by both clean and backdoored models to generate Grad-CAM heatmaps. Heatmap differences are then computed, features are extracted, and these features are fed into different classifiers.

F. Feature Extraction

We take a 30-dimensional feature vector from each Grad-CAM heatmap difference to measure how much more attention the clean model gets than the backdoored model. There are five groups of features that capture different aspects of attention change: statistical, spatial, magnitude-based, similarity-based, and gradient-based features.

Magnitude-based and statistical features tell us how strong the change in attention is, while spatial features tell us where the changes in attention happen. Features based on similarity look at how similar the two attention maps are, while features based on gradients look at changes in the structure of the heatmaps. Table I shows the full list of features that were taken out.

As examples, the mean attention difference and the ℓ_2 norm are computed as:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} D_i \tag{2}$$

$$\|D\|_2 = \sqrt{\sum_{i=1}^{N} D_i^2} \tag{3}$$

where D denotes the heatmap difference and N is the total number of pixels.

Each heatmap difference corresponds to one detection sample and is assigned a binary label:

- **0**: normal behavior (no backdoor detected), obtained from clean images
- **1**: backdoor behavior detected, obtained from triggered images

TABLE I: Summary of extracted features from Grad-CAM heatmap differences

Feature Category	Feature Names
Statistical	Mean, Standard Deviation, Maximum, Minimum, Median, 25th Percentile, 75th Percentile, Range
Spatial	Entropy, Center of Mass (X/Y), Spatial Variance (X/Y), Gini Coefficient, Center Shift Distance, Shift Direction
Magnitude	ℓ_1 Norm, ℓ_2 Norm, ℓ_∞ Norm, Percentage of Large Changes, Percentage of Small Changes, Energy Concentration
Similarity	Pearson Correlation, Cosine Similarity, Structural Similarity (SSIM), KL Divergence
Gradient	Gradient Magnitude Mean, Edge Density, Smoothness, Frequency Energy

This process creates a balanced dataset of 20,000 samples (10,000 normal and 10,000 backdoor) that is used for supervised detection.

G. Detection Algorithm

We set up backdoor detection as a binary classification task and tested a wide range of models to make sure they all worked well. We used both traditional ensemble learners (like XGBoost, LightGBM, Random Forest, Gradient Boosting, and AdaBoost) and Support Vector Machines (SVM) with an RBF kernel to set strong baselines for tabular data. To get the most stable predictions, we used a soft-voting ensemble that included Random Forest, XGBoost, and LightGBM. We also used deep learning architectures, like 2D Convolutional Neural Networks and spatial attention mechanisms, to find the high-dimensional dependencies of a backdoor.

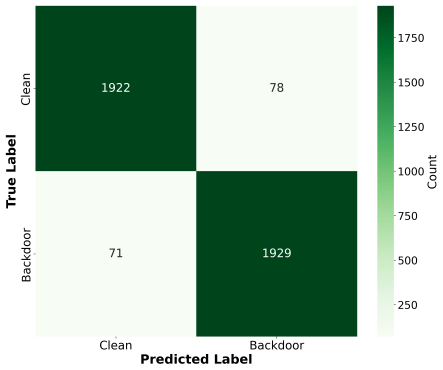

Fig. 2: Confusion matrix for test set data.

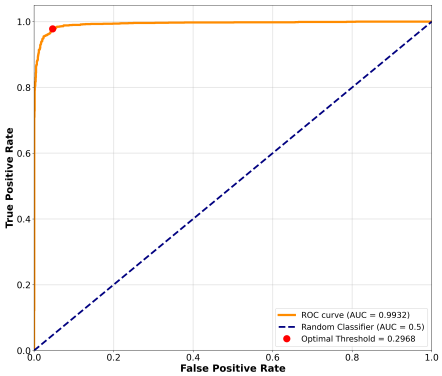

Fig. 3: ROC curve of the proposed backdoor detection method.

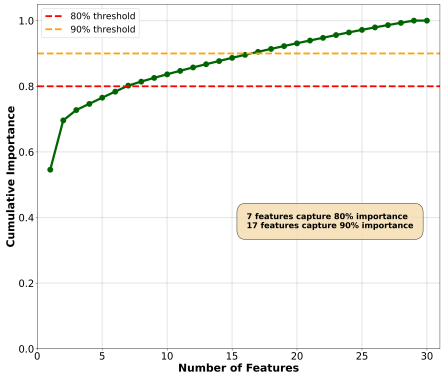

Fig. 4: Cumulative feature importance of the detector.

Given a feature vector $\mathbf{x}$, the detector predicts the class label as:

$$\hat{y} = \begin{cases} 1, & \text{if } f(\mathbf{x}) \geq \tau \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $f(\cdot)$ denotes the classifier model and τ is the decision threshold.

The dataset is divided into 70% training data and 30% testing data using a stratified strategy. To pick hyperparameters, we use grid search with cross-validation. The main metric for optimization is ROC-AUC.

H. Evaluation Metrics

Standard metrics, such as accuracy, precision, recall, F1-score, and ROC-AUC, are used to measure how well detection works. To lower the chance of missing a backdoored model, recall is stressed.

IV. Results and Discussions

In this part, we look at how well the proposed SIBA backdoor detection framework works. We present quantitative detection results and offer comprehensive analysis utilizing confusion matrices, ROC curves and feature importance.

A. Detection Performance and comparative analysis

Table II provides a thorough assessment of the proposed method in comparison to a variety of baseline algorithms, including conventional machine learning classifiers and deep neural networks. The results show a clear order of performance, with the proposed Spatial Attention mechanism getting the best scores on all metrics that were tested.

1) Baseline and Ensemble Analysis: The SVM with an RBF kernel was surprisingly strong among traditional machine learning models. It had a Test Accuracy of 96.54% and an AUC of 0.9931. This means that the backdoor patterns can be separated by a significant amount when they are projected into a higher-dimensional feature space. LightGBM (96.22% Test Acc) and XGBoost (96.05% Test Acc) are two tree-based gradient boosting methods that also did well. These models are great at working with tabular feature representations of inputs, but they usually can't capture the complex, local spatial dependencies that are part of trigger patterns as well as deep learning methods can. The Soft Voting Ensemble, which combines the predictions of several classifiers (Random Forest, XGBoost, and LightGBM), did the second-best job overall (Test F1: 97.68%). The ensemble lowered the chance of misclassifying ambiguous samples by averaging out the differences between individual models. This proved that decision-making based on consensus works for backdoor detection.

2) Deep Learning and Spatial Attention: The 2D CNN, which is the baseline for deep learning, did better than most traditional methods (Test Acc: 97.12The Spatial Attention method that was suggested fixes this problem and had the best overall performance, with a Test Accuracy of 98.34% and a ROC-AUC of 0.9978. The attention mechanism lets the model focus on the backdoor trigger's unique features while blocking out background noise by changing the importance of different input regions over time. This feature leads to a high recall rate (98.43%), which means that the model rarely misses a backdoored sample, and a high precision rate (98.25%), which means that it doesn't cause false alarms. Also, the difference in accuracy between training and testing for the Spatial Attention model is very small (about 0.8%), which shows that the model has learned strong features instead of just memorizing the noise from the training data.

B. Feature Importance Analysis

We look at the feature importance scores that our best classifier gives to see which features are most important in order to find backdoors. Table III shows how important the extracted features are. It is clear that features related to gradients are the most important for detection.

The gradient mean feature has the highest importance score, 0.5458, and the smoothness feature has the second highest score, 0.1501. These two things are a big part of how the model makes decisions. This means that backdoored models change attention in big, structured ways. Other features, like minimum value, variance along the Y-axis, SSIM, and Pearson correlation, give less important but still useful information.

TABLE II: Comprehensive Backdoor Detection Performance Comparison on Training and Test Sets

Method	Train Acc	Train Prec	Train Rec	Train F1	Train AUC	Test Acc	Test Prec	Test Rec	Test F1	Test AUC
XGBoost	0.9688	0.9680	0.9696	0.9688	0.9945	0.9605	0.9592	0.9618	0.9605	0.9918
Random Forest	0.9598	0.9610	0.9586	0.9598	0.9921	0.9561	0.9595	0.9527	0.9561	0.9899
Gradient Boosting	0.9642	0.9645	0.9639	0.9642	0.9932	0.9582	0.9565	0.9599	0.9582	0.9905
AdaBoost	0.9412	0.9385	0.9439	0.9412	0.9854	0.9388	0.9312	0.9464	0.9387	0.9841
LightGBM	0.9721	0.9715	0.9727	0.9721	0.9958	0.9622	0.9610	0.9634	0.9622	0.9925
SVM (RBF)	0.9745	0.9750	0.9740	0.9745	0.9962	0.9654	0.9675	0.9633	0.9654	0.9931
Soft Voting Ensemble	0.9872	0.9865	0.9879	0.9872	0.9985	0.9768	0.9755	0.9781	0.9768	0.9962
2D CNN	0.9814	0.9805	0.9823	0.9814	0.9978	0.9712	0.9705	0.9719	0.9712	0.9945
Spatial Attention	0.9915	0.9910	0.9920	0.9915	0.9991	0.9834	0.9825	0.9843	0.9834	0.9978

TABLE III: Top 10 most important features used for backdoor detection

Rank	Feature Name	Importance Score
1	Gradient Mean	0.5458
2	Smoothness	0.1501
3	Minimum Value	0.0315
4	Variance (Y-axis)	0.0189
5	SSIM	0.0188
6	Pearson Correlation	0.0185
7	75th Percentile (Q75)	0.0182
8	Median	0.0124
9	Standard Deviation	0.0112
10	Euclidean Distance	0.0110

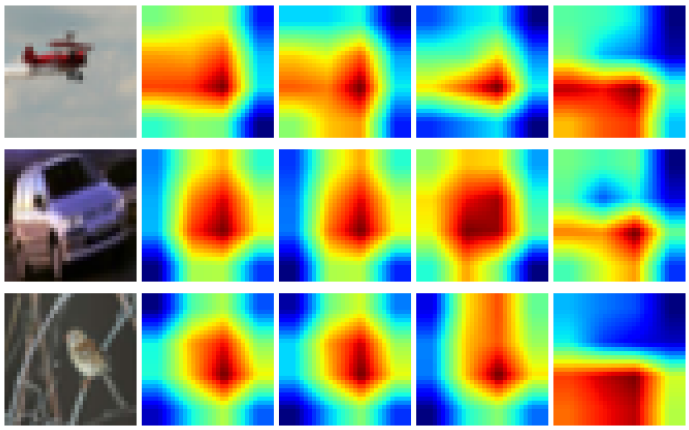

Fig. 5: Multi-sample Grad-CAM comparison across different classes. Each row corresponds to a different test image. From left to right, the columns show: (1) the clean input image, (2) clean model with clean input, (3) clean model with triggered input, (4) backdoored model with clean input, and (5) backdoored model with triggered input. The backdoored model shows clear attention shifts under triggered inputs, while the clean model maintains stable attention patterns.

Figure 4 shows the cumulative feature importance in more detail. The results show that only 7 features make up about 80% of the total importance, and 17 features make up about 90%. This shows that you can find backdoors with only a few features.

C. Visual Analysis of Grad-CAM Heatmaps

To visually analyze the behavior of the proposed detector, we compare Grad-CAM heatmaps from the clean model and the backdoored model under clean and triggered inputs.

Figure 5 shows multiple test samples from different classes. From left to right each column displays: (1) the clean input image, (2) clean model with clean input, (3) clean model with triggered input, (4) backdoored model with clean input, and (5) backdoored model with triggered input.

The clean model pays consistent and meaningful attention to both clean and triggered inputs. The backdoored model, on the other hand, shows clear changes in attention when the trigger is present, focusing on areas that aren't important. These patterns show up consistently across different samples, which shows that backdoor attacks cause changes in model attention that can be seen.

V. Discussion

The results show that looking at Grad-CAM attention patterns in clean and suspicious models is a good way to find backdoor attacks that are sparse and hard to see. This method looks at changes in the model's behavior instead of looking for the trigger directly. This works well even when the trigger is very small or hard to see.

In general, the proposed framework has a high detection rate and is easy to understand, making it a good way to find SIBA-style backdoor attacks.

VI. Conclusion

This paper introduces a straightforward and efficient technique for identifying Sparse and Invisible Backdoor Attacks (SIBA) in deep neural networks. The suggested method employs gradient-based features and different classifiers, without visible triggers.

The experimental results show that both the training and test data have a high detection rate. Feature analysis shows that only a few gradient features are needed to find SIBA attacks, even when methods like Grad-CAM don't work.

Scope and Limitations

This work focuses on a targeted study of Sparse and Invisible Backdoor Attacks (SIBA) under a controlled evaluation setting. Experiments are conducted on CIFAR-10 using different architectures to isolate and analyze attention-based behavioral signatures introduced by SIBA. While broader evaluation across datasets and attack types is important, such extensive benchmarking is beyond the scope of this work and is left for future study.

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